

Ex. No.: 6b)

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SHORTEST JOB FIRST

Aim:

To implement the Shortest Job First (SJF) scheduling technique

Algorithm:

1. Declare the structure and its elements.
2. Get number of processes as input from the user.
3. Read the process name, arrival time and burst time.
4. Initialize waiting time, turnaround time & flag of read processes to zero.
5. Sort based on burst time of all processes in ascending order.
6. Calculate the waiting time and turnaround time for each process.
7. Calculate the average waiting time and average turnaround time.
8. Display the results.

Program Code:

```
#include <stdio.h>
```

```
struct process {  
    char name[10];
```

```
    int arrival-t, burst-t, waiting-t, turnaround-t,  
        pid;
```

```
int main() {
```

```
    int n, i, j;
```

```
    float total-wt = 0; total-tat = 0;
```

```
    printf("Enter the no. of processes:");
```

```
    scanf("%d", &n);
```

```
    struct process p[n];
```

```
    printf("Enter the Burst Time for each Process:\n");
```

```

for (i=0; i<n; i++) {
    sprintf (p[i].name, "p%d", i+1);
    scanf ("%d", &p[i].burst-time);
    p[i].waiting-t = 0;
    p[i].turnaround-t = 0;
}

```

```

for (i=0; i<n-1; i++) {
    for (j=i+1; j<n; j++) {
        if (p[i].burst-t > p[j].burst-t) {
            struct Process temp = p[i];
            p[i] = p[j];
            p[j] = temp;
        }
    }
}

```

```

int current-t = 0;
for (i=0; i<n; i++) {
    p[i].waiting-t = current-t;
    p[i].turnaround-t = p[i].waiting-t + p[i].burst-t;
    total-wt += p[i].waiting-t;
    total-tat += p[i].turnaround-t;
    current-t += p[i].burst-t;
}

```

```

printf ("\n Process\t Burst Time\t waiting Time\t\n");
printf ("Turnaround Time\n");

```

```

for (i=0; i<n; i++) {
    printf ("%s\t %d\t %d\t %d\t %d\n", p[i].name,
        p[i].burst-t, p[i].waiting-t, p[i].turnaround-t);
}

```

}

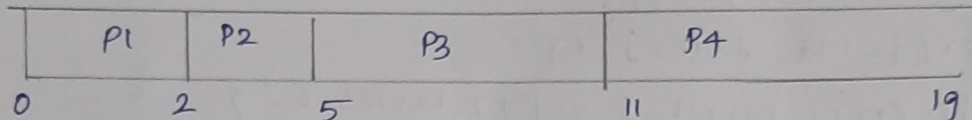

```

printf (" \n Average waiting time: %.2f", total-wt/n);
printf (" \n Average Turnaround time: %.2f", total-tat/n);

return 0;
}

```

Gantt chart:

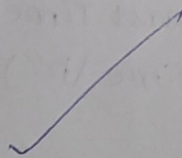


TABULATION: consider AT as 0.

Process	Burst Time (ms)	waiting time (ms) (TAT-BT)	Turnaround time = (CT-AT) (ms)	Completion time (ms)
P1	2	0	2	2
P2	3	2	5	5
P3	6	5	11	11
P4	8	11	19	19

Average TAT = 9.25 ms

Average WT = 4.5 ms



Sample Output:

Enter the number of process:

4

Enter the burst time of the processes:

8 4 9 5

Process	Burst Time	Waiting Time	Turn Around Time
2	4	0	4
4	5	4	9
1	8	9	17
3	9	17	26

Average waiting time is: 7.5

Average Turn Around Time is: 13.0

Enter the number of processes: 4
Enter burst time of the process: 6 2 8 3

Process	Burst time	waiting time	Turnaround time
P2	2	0	2
P4	3	2	5
P1	6	5	11
P3	8	11	19

Average waiting time : 4.50

Average Turnaround Time : 9.25

Result:

Thus the code to implement the Shortest Job First (SJF) scheduling technique has been executed successfully.

